

Is Migration a source of “Global Socialization”?

Evidence from the MENA region

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September 19, 2026

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Abstract

Societies in North Africa and the Middle East are generally characterized as socially conservative, yet they witness a high degree of emigration to culturally dissimilar societies, begging the question whether migrant corridors contribute to global socialization in the MENA context. I create a subnational migration matrix based on the Facebook Social Connections Index and standardize Arab Barometer survey data, leveraging subnational variation in migration to study relationships between migration links and cultural norms, focusing on gender norms. Findings confirm that the migration matrix is informative about actual emigrant networks, and reveal that destination regions are statistically significantly related to regional gender norms.

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I would like to thank Valeria Churango and Jakhongir Sadiev for research assistance; the Harvard Mellon Urban Initiative and Mahindra Humanities Center for research funding; the Harvard Center for International Development for research support; Gordon Hanson, Gabriel Kreindler, and Rema Hanna for advising; and participants of the Harvard Graduate Student Workshop in Economic Development and the Harvard Kennedy School Graduate Student Workshop in Economic and Social Policy for helpful feedback and encouragement. I introduced the subnational migrant matrix in a working paper presented at the 19th ADB-BOK-IDB-Y-RISE(Yale)-Yonsei IPHC International Conference on Migration and Development in April 2026.

1 Introduction

Migration networks trigger international cultural flows in addition to flows of labor and financial funds, creating conditions conducive to the spread of norms. This paper considers international migration as a means of cultural transformation, and investigates the relationship between migrant networks and norms from the perspective of migrant-sending countries in the Middle East and North Africa. While these societies are often characterized as socially conservative regarding gender norms or religious pluralism, they witness high degrees of labor emigration to culturally divergent societies, leading to cultural exposure to both more progressive (e.g. European and North American) and more conservative (e.g. Gulf) host-country societies. I study the relationship between within-country variation in migrant networks and norms, creating a social media-based migrant matrix and leveraging a decade’s worth of public opinion surveys from the Arab world. I find that migrant networks are not only related to migration motives and prospective destinations, but also to gender norms.

Migration could impact culture in several ways. Rapoport, Sardoschau, and Silve (2021) theorize compositional change and cultural remittances as possible means, and argue that cultural remittance best explain the cultural convergence they observe between country pairs using World Values Survey Data and UN migration estimates. Tuccio and Wahba (2018) contend that return migrants assimilate to destination culture, demonstrating that Jordanian return migrants from the Gulf have more conservative gender norms than non-migrants after accounting for migrant selection.

Fertility norms have been studied by demographers and economists as a prime example of norms spreading and changing through migrant networks. Daudin, Franck, and Rapoport (2019) and Melki et al. (2024) identify migration as a means by which the fertility transition spread historically within France and Europe, while Godlonton and Theoharides (2025) create an emigrant shift-share instrument for the Philippines to argue a similar dynamic is playing out in contemporary Philippine regions. The demographer Fargues (2006) posits that migrants disseminate destination country fertility norms regardless of whether norms are for many or few children, and compares North African countries to show fertility norm transmission through migrant networks from destinations with higher and lower fertility rates. Beine, Docquier, and Schiff (2013) run cross-country regressions supporting his hypothesis, and Tuccio and Wahba (2018) uncovers that Jordanian return migrants from the Gulf, an area with higher fertility norms, have larger families. The migrant-fertility norm transmission literature reveals that migration can serve as a conduit for “liberal” or “conservative” norms, depending on migrant destination country.

I contribute to the literature by examining subnational variation in migration destinations and emigration rates, similar to Fargues (2006), Barsbai et al. (2017), and Godlonton and Theoharides (2025), across a host of origin countries by leveraging a novel subnational migrant matrix based on social media data. My methodology builds on economic literature estimating migrant stocks, incorporating social media-based data to create a global subnational migrant matrix covering 67% of the 2020 global migrant stock (UN 2020). The methodology for my migrant matrix resembles Nunn (2008)’s estimation of forced migration flows during the African slave trade, and I build on a literature leveraging the Facebook Social Connections Index for migration and social science research (Bailey et al. 2018; Bonaudio et al. 2024; Mooers 2026). I identify global migrant stocks by subnational origin and

destination regions. My matrix includes countries covered by the Facebook Social Connections Index and UN bilateral migration matrix, and I derive emigrant stock estimates at the GADM1 level ($\approx$ US states) for most migrant-sending countries, and at the NUTS3 level for European destinations ($\approx$ US counties), based on imputed Facebook connections.

This paper proceeds to describe the data and empirical strategy, including the methodology used to create the subnational migrant matrix. It then goes over initial results, and concludes.

2 Data

I analyze spatial datasets, migration estimates, social media-based data, and public opinion survey data to document regional variation in migration networks and cultural norms. I pool national survey microdata across six waves of Arab Barometer surveys to estimate regional variation in gender norms, following a common social science practice of pooling surveys to improve sub-group analysis. Data sources include:

Arab Barometer Survey Data: nationally representative public opinion survey microdata from Arab Barometer waves II to VII covers countries across North Africa and the Middle East between 2010 to 2022 (Arab Barometer 2009–2024). Surveys include Likert-style questions about social, political and religious views in addition to demographic information. Waves II to VII record respondents’ region of residence ($\approx$ US state), allowing me to compute regional mean values after standardizing question encodings across surveys and pooling results by regions.

Facebook Data: I estimate social media links in October 2021 using the Facebook Social Connections Index (SCI) and Facebook usership statistics. The Facebook Social Connections Index (SCI), based on active Facebook links in October 2021, effectively allows one to compare the likelihood of a Facebook friendship occurring between any two regions in the world Bailey et al. (2021b) and Bailey et al. (2021a). I also consult country-level Facebook usership statistics for October 2021 (NapoleonCat 2025), based on Facebook advertising.

Global Human Settlement Layer (GHSL): the Global Human Settlement Layer (GHSL) provides planet-wide shapefiles with 5 year population estimates in 1km grid cells, based on administrative data and satellite-imagery (Schiavina, Freire, and MacManus 2023). Subnational populations in 2020 are computed from these shapefiles, defining regions using international subnational administrative boundaries from GADM version 2.8 (GADM 2015a; GADM 2015b) and NUTS boundaries from the EU (European Commission, Eurostat (ESTAT), GISCO 2019; European Commission, Eurostat (ESTAT), GISCO 2024).

International Migration Data: the United Nations estimates a global migrant matrix with 5 year estimates of binational migrant stocks (UN 2020), and I allocate 2020 migration stock estimates to subnational origin and destination regions based on imputed Facebook links. I also consult Spanish population registry data to validate the global subnational migrant matrix (INE 2023).

3 Empirical Strategy

3.1 Subnational Migrant Matrix

A major contribution of my paper is estimating a subnational migrant matrix covering the majority of global migration. Binational migration estimates, recording migrant stocks by origin and destination country (e.g. Moroccans in the United States) are available from the United Nations (UN 2020), but estimates at the subnational origin and destination level (e.g. the number of Moroccans from Casablanca living in the state of Massachusetts) are less common. While current subnational migrant matrices are constrained in examining a single migration corridor (Caballero, Cadena, and Kovak 2023), destination country (Melki et al. 2024; Salem and Seck 2025), or origin country (Khanna et al. 2022; Godlonton and Theocharides 2025), I leverage social media-based data to estimate a subnational migrant matrix covering 67% of world migrants in 2020 (UN 2020). While I focus on emigration from the Middle East and North Africa, my methodology would apply to other regions such as Latin America and Sub-Saharan Africa.

My estimation strategy builds upon Bailey et al. (2018)’s remark that social media connections are related to migration, and bears similarities to Nunn (2008)’s approach to estimate national exposure to the African slave trade. Through an analysis of an earlier version of the Social Connections Index, Bailey et al. (2018) report a strong relationship between international immigration and social media connections between foreign countries and US counties. They report statistically significant positive relationships between a county’s ancestry or birth in a foreign country and contemporary social connections index between the US county and foreign country. Nunn (2008) brings together multiple data sources to estimate country-level slave exports from slave ethnicity subsamples and broader shipping data reporting total slave exports by African port or region. He collects data on the ethnic distribution of enslaved persons within slave trades using 54 distinct samples covering 80,656 slaves and 229 ethnicities. He multiplies the ethnic shares within each slave trade from these samples by total flows in each slave trade route to estimate slave flows by ethnicity, and after assigning ethnic groups across countries, sums up the number of slaves by country. His methodology rhymes with mine, although I use inferred social media links to estimate shares of migrants within subnational corridors, and multiply shares of links by national migrant stocks to obtain a subnational emigrant matrix. There are stark contrasts as I examine contemporary and mostly voluntary migration rather than historical coerced migration.

I estimate subnational migrant stocks based on subnational shares of imputed Facebook connections and UN migration estimates. Facebook is a promising source for data on social networks due to its widespread international use. Facebook advertising reached 2.28 billion users globally in October 2021 (Kemp 2021).

To create a migrant matrix, I use Facebook Social Connections Index (SCI) data from Meta (Bailey et al. 2021b; Bailey et al. 2021a), social media statistics regarding Facebook usership from NapoleonCat (2025), and a binational migrant matrix from the UN which provides five-year estimates of migrant stocks by origin and destination country (UN 2020). Let i be a region in migrant origin country o , and j a region in migrant destination country d . The social connections index between any two regions i and j ($SCI_{i,j}$) is a scaled version of the following fraction: the number of active Facebook friendships between users in i and

j divided by the product of the number of Facebook users in both regions, calculated based on Facebook usership in October 2021:

$$SCI_{i,j} \approx \frac{Connections_{i,j}}{Users_i \cdot Users_j} \quad (1)$$

I first estimate regional Facebook usership to back out Facebook connections from SCI. I estimate regional usership by multiplying regional populations by national Facebook usage rates. My strategy bears some similarity to Mooers (2026), who estimates cross-county friendships in the United States from the SCI by using total county population. I estimate regional Facebook users rather than merely using population, which I consider more appropriate in my context given cross-country variation in Facebook usage rates. I estimate regional population, $Pop_{i,2020}$, from Global Human Settlement Layer 2020 shapefile population estimates within GADM and NUTS3 boundaries (Schiavina, Freire, and MacManus 2023; GADM 2015b; European Commission, Eurostat (ESTAT), GISCO 2019). I estimate national Facebook usage rates, $FBRate_{o,2021}$ by dividing the number of Facebook users in country o in October 2021 from NapoleonCat (which derives its numbers from Facebook advertising) by national population estimates in 2020 according to GHSL (NapoleonCat 2025; Carioli et al. 2024), capping this rate at 1. I then multiply region i 's population by its country o 's Facebook usage rate to estimate the number of Facebook users in subnational region:

$$\widehat{Users_{i,o,2021}} = Pop_{i,2020} \cdot FBRate_{o,2021} \quad (2)$$

Next, I back out total scaled connections between two regions (region i in country o and region j in country d) by multiplying their SCI ($SCI_{i,j}$) by estimates of each regions' Facebook usership in 2020:

$$\widehat{Connections_{i,j,2021}} = SCI_{i,j,2021} \cdot \widehat{Users_{i,o,2021}} \cdot \widehat{Users_{j,d,2021}} \quad (3)$$

I then estimate subnational migrant stocks by allocating national migrant stocks to regional corridors based on their share of binational Facebook connections. I compute a subnational regional corridor's share of total connections within the national corridor, and estimate subnational emigrant stocks by multiplying UN national migration stocks with this share:

$$\widehat{Emigr_{i,o,j,d,2020}} = EmigrUN_{o,d,2020} \cdot \frac{\widehat{Connections_{i,j,2021}}}{\sum_i^O \sum_j^D \widehat{Connections_{i,j,2021}}} \quad (4)$$

The central assumptions of my subnational migration matrix are (1) that Facebook usership rates are constant across subnational regions within countries, and (2) that shares of binational social media links correspond to subnational shares of binational migrants. Anderson and Auxier (2021) provide evidence in favor of assumption (1), as they report nearly identical rates of Facebook usage between urban and suburban Americans (70%) and rural Americans (67%) in early 2021. Bailey et al. (2018) provide suggestive evidence in favor of assumption (2) as they find the SCI between US counties and foreign countries is significantly related to immigration, proxied by birth country.

To test assumption (2), I compare the geographic distribution of Facebook links and of immigrants in Spain in 2021 by country of origin, using Spanish population registry data for 2021 (INE 2023). Figure 1 reveals that social media links and migrant stocks are highly correlated, with a correlation coefficient of 0.914 between regional shares of binational Facebook links and shares of immigrant stocks from the same country.

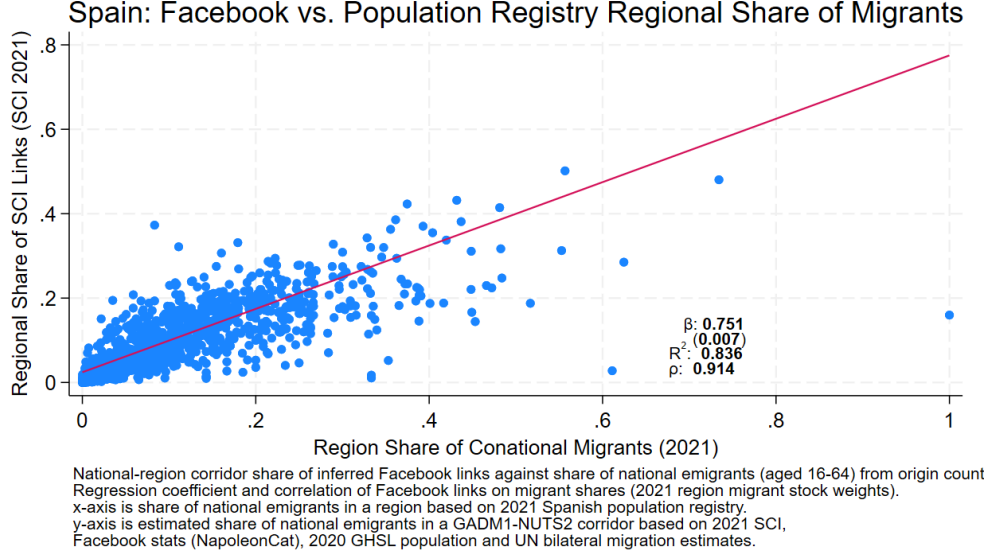


Figure 1: SCI vs. Population Registry Shares (2021)

3.2 Descriptive Regressions

To investigate the relationship between emigration networks and norms, I run descriptive regressions leveraging within-country variation in emigration. Two dimensions are of particular interest when considering cultural exchange: scale and destination. To account for population differences, I compute emigration rates, dividing emigrant stocks by regional populations, and use these as the explanatory variable.

My theory is that migrant networks serve as cultural conduits between destination and origin regions. This means that migrant destination matters. To simplify matters, I compute emigration rates $emrate_{i,o,D}$ by set of destination regions D , summing up emigrant stocks from region i living in all regions j in a set of destinations D . This implicitly assumes homogenous culture within destination D , but allows for cultural transmission impacts to differ across destinations groupings. I compute emigration rates by destination country, and by larger country groupings, namely migration to North America and Europe (“West”) and emigration to the United Arab Emirates, Bahrain, Kuwait, Oman, Qatar, and Saudi Arabia (“Gulf”).

$$emrate_{i,o,D} = \frac{\sum_j^{J \subset D} Emigr_{i,o,j,2020}}{Pop_{i,2020}} = \frac{Emigr_{i,D,2020}}{Pop_{i,2020}} \quad (5)$$

My outcome variable is regional mean agreement (or disagreement) with standardized survey questions. I convert Likert scales into Boolean variables indicating agreement (or disagreement) with a statement, standardizing similar questions across survey waves. I then compute regional mean agreement (or disagreement) with the statement, and regress these regional means against emigration rates $emrate_{i,o,d}$, including origin country fixed-effects γ_o so regression coefficients are identified based on within-country variation in emigration rates. This removes time-invariant, country-specific characteristics which could otherwise provide omitted variable bias, such as a common colonial history that might shape modern values and migration networks. To account for additional sources of omitted variable bias, I can add regional controls $\mathbf{X}_{i,o}$.

$$proValue_{i,o} = emrate_{i,o,D} + \gamma_o + \epsilon_{i,o} \quad (6)$$

$$proValue_{i,o} = emrate_{i,o,D} + \gamma_o + \mathbf{X}_{i,o} + \epsilon_{i,o} \quad (7)$$

These regressions are inherently descriptive and not causal. The data provides a snapshot of subnational migrant networks and norms around 2020, and does not permit an analysis of changes in migration networks or norms over time. Docquier, Tansel, and Turati (2020) document cultural self-selection into migration among Middle Eastern and North African migrants, so there could be reverse causality if regional emigration networks are the result of pre-existing cultural differences, for example if more cosmopolitan locations send more emigrants to cosmopolitan destinations. While I cannot rule out this concern, I address concerns of reverse causality by controlling for urbanization, since urban locations are often less conservative.

4 Results

I run regional regressions against emigration rates and networks. Before focusing on gender norms, I confirm that migrant networks, as estimated by the subnational migrant matrix, are associated with migration-related variables in the way one would expect. After this reality check, I proceed to examine relationships between cultural norms and emigration, and provide suggestive evidence that migrant networks beget cultural remittances, as I find that emigrant networks are statistically significantly associated with relative progressivism on gender norms.

4.1 MENA Reality Check

I first validate that my subnational migrant matrix successfully proxies emigrant networks in the Middle East and North Africa. Multiple Arab Barometer waves include questions on migration intentions, migration motives, and prospective migrant destinations. I regress regional mean migration intentions and motives on total emigration rates, and prospective destinations on country-specific emigration. Table 1 reveals that higher emigration rates are significantly associated with increased intention to emigrate and motivation for family reasons. This suggests that social media-based migration matrix is measuring total emigration, since people in places with more emigrants would be more likely to have family members abroad. Furthermore, prospective destinations are significantly related to emigration rates for six out of eight countries I investigate in Table 2. I interpret this as evidence that the subnational migration network accurately identifies variation in migrant destination countries within sending regions.

These results strongly suggest that the subnational migration matrix accurately represents existing subnational emigrant networks for Middle Eastern and North African countries. Having demonstrated their validity in this context, I proceed to investigate relationships between emigrant networks and cultural norms.

4.2 Migration and Culture

I investigate the relationship between migrant networks and gender norms, using standardized Arab Barometer survey responses on questions related to gender attitudes. The questions I consider are related to male preference in politics and higher education, household decision making, and female employment. I regress regional means against two broad emigration rates, namely emigration to Western countries (North America and Europe), generally considered progressive on gender issues, and Gulf countries (United Arab Emirates, Bahrain, Kuwait, Oman, Qatar, and Saudi Arabia), generally considered conservative on gender issues. I include regional urbanization rates as a control, based on the Arab Barometer, and include country fixed effects. I find that emigration to Western countries is statistically significantly related to more progressive gender views on higher education and female employment outside the home. A one percentage point increase in a region's emigration rate to North American and European destinations is associated with a 0.7 percentage point decline in a region's share of respondents agreeing that university education is more important for men than for women, holding country and urban population share fixed, a relationship

Table 1: MENA Migration Motives

	(1)	(2)	(3)
	Would Migrate	For Family	For Economics
Emigr Rate	0.401** (0.203)	0.642*** (0.186)	-0.0719 (0.391)
Country FE	X	X	X
N	162	159	159
N Countries	8	8	8
Adj. R2	0.720	0.650	0.663
Mean	0.266	0.039	0.759

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Arab Barometer, UN, Facebook, GHSL, GADM, EU, NapoleonCat.

Region-level, population-weighted regressions with country fixed effects, robust standard errors.

Table 2: MENA Migrant Intended Destinations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	France	Spain	Italy	UK	USA	Germany	Saudi	UAE
Emigr Rate in Destination	0.780*** (0.181)	1.519*** (0.523)	1.080*** (0.308)	12.18*** (4.056)	3.534*** (0.820)	2.533** (1.238)	0.797 (0.815)	0.194 (1.133)
Country FE	X	X	X	X	X	X	X	X
N	162	162	162	162	162	162	162	162
N Countries	8	8	8	8	8	8	8	8
Adj. R2	0.657	0.790	0.590	0.382	0.679	0.553	0.573	0.485
Mean	0.045	0.024	0.034	0.012	0.033	0.035	0.027	0.031

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Arab Barometer, UN, Facebook, GHSL, GADM, EU, NapoleonCat.

Region-level, population-weighted regressions with country fixed effects, robust standard errors.

Table 3: MENA Emigration and Regional Gender Norms

	(1) Men better Politicians	(2) Men priority for University	(3) Men Final Say Family Decisions	(4) Women may Work Outside
Emigr Rate to West	-0.129 (0.206)	-0.714*** (0.176)	-0.193 (0.281)	0.519* (0.281)
Emigr Rate to Gulf	-1.069 (1.259)	0.365 (0.940)	1.792* (0.969)	-0.327 (0.872)
Urban Share	-0.0511 (0.0337)	-0.0751*** (0.0190)	-0.0887*** (0.0287)	0.0592* (0.0314)
Country FE	X	X	X	X
N	162	162	162	133
N Countries	8	8	8	6
Adj. R2	0.760	0.355	0.719	0.190
Mean	0.680	0.219	0.622	0.811
p-val West=Gulf	0.475	0.274	0.058	0.367

Standard errors in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Arab Barometer, UN, Facebook, GHSL, GADM, EU, NapoleonCat.

Region-level, population-weighted regressions with country fixed effects, robust standard errors.

which is statistically significant at the 1% level. Emigration to the West is statistically significantly related to increased agreement that women may work outside the house, although the relationship is weaker. However, I cannot reject the null hypothesis that coefficients on emigration rates to the West and the Gulf are equal. Emigration to Gulf countries is slightly significantly related to more conservative views on household decision making, specifically agreeing that husbands should have the final say in decisions concerning the family. This positive coefficient is statistically different from the coefficient on emigration to the West, at least at the 10% level (p-value of 0.058). These findings that emigrant networks are related to regional variation in gender attitudes suggest that gender norms in sending regions are impacted by those in emigrant destinations.

5 Conclusion

In conclusion, I create a global subnational migrant matrix based on social media links, and provide evidence suggesting that emigrant networks, in particular migrant destinations, impact gender norms in migrant sending countries in the Middle East and North Africa. I validate the matrix in the MENA context using Arab Barometer survey responses on migration intentions and prospective destinations, and proceed to investigate relationships between migrant networks and gender norms using standardized Arab Barometer survey data. I find that migration to Western countries is associated with more liberal gender attitudes towards female higher education and employment outside the home, while migration to Gulf countries is associated with more conservative gender attitudes towards household decision making. In the near future, I would extend gender norm outcomes to investigate reported behavior as well as attitudes, and leverage the breadth of Arab Barometer survey to investigate additional cultural norms. Future work would incorporate additional survey data, including the World Values Survey, and extend to additional world regions such as Latin America and Sub-Saharan Africa using regional Latinobarometer and Afrobarometer survey data.

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